

BRACKNELL BEAUFORT, United Kingdom

WMO: 037630

Lat: 51.383N

Lon: 0.783W

Elev: 74

StdP: 100.44

Time Zone: 0.00 (EUW)

Period: 94-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.1	-2.8	-6.9	2.1	-1.0	-5.4	2.4	-0.5	7.9	6.8	7.1	7.2	0.9	50	0.503

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.2	27.5	18.4	25.2	17.4	23.4	16.8	19.5	24.7	18.5	23.4	17.5	21.9	2.6	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.6	12.7	21.7	16.7	12.0	20.4	15.8	11.3	19.2	55.9	24.4	52.4	23.4	49.5	22.1	23.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.8	6.0	5.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			-7.1	20.9	1.3	1.3	-8.0	21.8	-8.8	22.6	-9.6	23.3	-10.5	24.3	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.3	4.5	5.6	7.0	8.5	11.7	14.6	16.9	17.3	13.9	11.1	7.9	4.8	(6)
(7)		DBStd	5.33	3.47	3.21	2.76	3.21	2.99	2.52	2.38	2.83	2.23	3.01	3.18	3.91	(7)
(8)		HDD10.0	742	171	125	99	65	17	1	0	0	1	23	76	165	(8)
(9)		HDD18.3	2956	428	357	352	295	206	116	58	55	134	224	312	419	(9)
(10)		CDD10.0	867	1	1	6	20	69	137	213	226	118	57	14	4	(10)
(11)		CDD18.3	39	0	0	0	0	0	3	13	23	1	0	0	0	(11)
(12)		CDH23.3	418	0	0	0	1	12	49	130	218	8	0	0	0	(12)
(13)		CDH26.7	88	0	0	0	0	0	9	22	58	0	0	0	0	(13)
(14)	Wind	WSAvg	2.1	2.5	2.7	2.5	2.3	2.2	2.1	1.8	1.6	1.8	2.1	2.0	2.3	(14)
(15)	Precipitation	PrecAvg	708	75	53	43	53	48	51	49	58	57	74	80	69	(15)
(16)		PrecMax	949	184	128	113	141	105	128	122	114	116	154	159	130	(16)
(17)		PrecMin	524	11	4	7	3	4	13	12	9	10	29	30	13	(17)
(18)		PrecStd	114	37	33	23	34	25	31	24	33	29	34	39	34	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.4	13.2	16.3	20.9	24.8	28.2	29.3	31.2	24.7	21.0	15.6	13.6	(19)
(20)			MCWB	10.6	10.0	10.6	13.3	15.3	17.8	19.3	20.2	17.8	16.5	13.6	11.9	(20)
(21)		2%	DB	11.6	12.2	14.4	18.1	22.8	24.6	26.7	28.2	21.2	17.9	14.4	12.7	(21)
(22)			MCWB	9.8	9.7	9.7	11.5	15.6	16.3	18.0	18.6	15.9	14.3	13.1	11.2	(22)
(23)		5%	DB	10.6	11.2	13.0	15.9	20.2	22.2	24.6	25.8	19.7	16.7	13.5	11.9	(23)
(24)			MCWB	9.0	9.1	9.5	10.6	14.3	15.7	17.6	18.2	14.9	13.9	12.1	10.7	(24)
(25)		10%	DB	9.7	10.2	11.7	14.0	17.5	20.4	22.9	23.3	18.2	15.6	12.6	10.9	(25)
(26)			MCWB	8.3	8.3	8.9	9.8	12.8	14.7	16.8	17.2	14.3	13.6	11.1	9.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.2	11.1	12.1	13.9	17.9	18.9	21.1	21.2	18.1	17.1	14.1	12.5	(27)
(28)			MADB	12.3	12.7	14.6	17.5	22.1	26.2	27.2	28.7	22.3	20.2	14.8	13.1	(28)
(29)		2%	WB	10.1	10.2	11.0	12.4	16.2	17.3	19.5	19.9	16.9	15.5	13.4	11.7	(29)
(30)			MADB	11.2	11.6	12.8	16.0	21.1	22.5	24.3	25.2	20.2	17.0	14.2	12.5	(30)
(31)		5%	WB	9.2	9.5	10.3	11.4	14.6	16.4	18.3	18.8	16.1	14.6	12.4	10.7	(31)
(32)			MADB	10.3	10.9	12.0	14.6	19.3	20.9	23.0	23.6	18.5	16.0	13.4	11.7	(32)
(33)		10%	WB	8.4	8.6	9.5	10.4	13.3	15.5	17.2	17.8	15.2	13.8	11.3	9.7	(33)
(34)			MADB	9.5	10.0	11.2	13.4	17.0	19.3	21.7	22.2	17.3	15.4	12.2	10.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.4	6.3	7.6	8.8	9.2	9.5	9.8	10.2	8.3	7.4	5.8	5.0	(35)
(36)			MCDBR	5.7	6.2	9.8	11.5	14.4	13.7	13.7	14.4	10.6	8.0	5.0	4.9	(36)
(37)			MCWBR	5.1	5.0	6.2	6.1	7.9	7.0	6.7	6.0	5.7	5.2	4.1	4.8	(37)
(38)		5% WB	MCDBR	5.8	6.1	7.3	9.9	13.0	11.6	11.4	12.1	9.0	7.2	4.6	4.7	(38)
(39)			MCWBR	5.5	5.5	5.5	6.1	7.7	6.4	6.1	5.7	5.5	5.5	4.4	4.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.329	0.345	0.381	0.403	0.406	0.405	0.400	0.391	0.374	0.358	0.332	0.318	(40)	
(41)		taud	2.365	2.371	2.265	2.224	2.255	2.284	2.307	2.349	2.391	2.418	2.409	2.351	(41)	
(42)		Ebn at Noon	677	774	806	833	848	851	849	837	809	748	670	623	(42)	
(43)		Edh at Noon	64	82	109	127	129	126	121	111	95	77	60	55	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.82	1.48	2.58	3.99	4.84	5.35	5.07	4.22	3.17	1.84	0.98	0.64	(44)	
(45)		RadStd	0.07	0.19	0.35	0.45	0.30	0.54	0.49	0.34	0.27	0.16	0.11	0.08	(45)	

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
				99% DB	99% DP	1% DB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (81 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page